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STUDY PROTOCOL

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Soft-tissue quadriceps tendon autograft during primary anterior cruciate ligament reconstruction in Skeletally-immature patients vs. Hamstrings: A protocol for a multi-centre randomized controlled trial

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Abstract

Background ACL tears in skeletally-immature patients are often treated surgically using soft-tissue autografts to offset increased risks of cartilage, meniscus and physeal injuries. Despite technological innovation, there remains a high failure rate among active young patients. We present a randomized controlled trial (RCT) to evaluate the soft-tissue quadriceps autograft compared to the standard hamstring autograft to treat primary ACL tears in the pediatric population.

Methods This international RCT will evaluate 352 skeletally-immature patients between the ages of 10–18 years undergoing primary ACL reconstruction to compare the effect of hamstring versus soft-tissue quadriceps autografts on ACL failure rate as a primary outcome, and return to sport, knee function, knee pain, health-related quality of life and health utility, range of motion and stability, and other adverse events at 24 months as secondary outcomes. Follow-up will occur at 6 weeks and 6, 12, 18, and 24 months postoperatively.

Discussion This trial will inform the optimal primary surgical management for ACL injuries as it pertains to all soft-tissue graft selection, to not only reduce the relatively high failure rates, but also improve the function and return-to-sports in this young, active population.

Trial registration This trial was registered on ClinicalTrials.gov (NCT03896464) on March 27th, 2019.

Keywords Anterior cruciate ligament reconstruction, Pediatric, Knee, Quadriceps, Hamstring, Skeletally-immature, Graft selection, Physeal, Protocol, Randomized controlled trial

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Background

Injuries to the anterior cruciate ligament (ACL) occur at annual frequencies ranging from 100,000 to 200,000 and have an annual economic impact exceeding $1 Billion US [1–5]. In pediatric patients, there has been a near exponential rise in the incidence of ACL injuries – possibly due to a combination of increased sports participation/specialization, and a heightened awareness/improvement in the detection of injuries [6, 7]. ACL injuries are most often treated surgically as young patients undergoing delayed ACL reconstruction after failed conservative measures often present with much higher incidences of associated meniscal and/or chondral pathology [8–12]. As a result, early ACL reconstruction to restore stability and facilitate a return to activities requiring pivoting and cutting/lateral movement has become the preferred treatment, demonstrating largely satisfactory outcomes for the majority (i.e. 75–97%) of patients [13–15].

However, despite continued technique and technological innovation, surgical reconstruction failure rates among active pediatric patients approximate 10–25%; with this particular population being 15 times more likely to re-injure their graft within the first postoperative year [16–19]. Moreover, though approximately 84% return to the same sport postoperatively, emerging data reveal less than ideal rates of return to preinjury level of activity (55–83%) particularly in the young athlete population (63–67%) [20–25]. Broadly speaking, failures of primary ACL reconstruction can be categorized according to one or a combination of traumatic, technical, and/or biologic causes [25–27]. Among these, the optimal selection and effect of graft choice on ACL reconstruction outcomes, for both adult and pediatric patients, have long been a matter of intense controversy [28–31]. The presence of open physes in the pediatric population specifically, presents several technical challenges in an effort to avoid growth arrest, leg-length discrepancy, and/or angular deformity from iatrogenic physeal violation. As a result, it is recommended that pediatric ACL reconstruction only use soft-tissue autografts (not allografts) [31, 32].

Currently, a hamstring autograft is the reference soft-tissue standard for primary pediatric ACL reconstruction. The hamstring tendons are advocated for their apparent reproducibility, applicability, low donor site morbidity (i.e. less patellofemoral pain and postoperative stiffness), and preservation of extensor mechanism strength [33]. However, more recent research suggests that the hamstring tendon may not be the most ideal in some cases, potentially due to its tendency to stretch over time, associated unpredictable intraoperative graft lengths and diameters, cutaneous sensory loss at the donor site, injury to neurovascular structures, and/or bone tunnel osteolysis [34–39]. Within the adult ACL population, there is a renewed interest in the biomechanical and biologic potential of the all-soft-tissue quadriceps tendon autograft; possibly due a combination of its increased tensile strength, predictable preoperative templating, less kneeling pain, less graft site donor morbidity, and/or comparable clinical outcomes [12, 40].

To date, the use of the quadriceps tendon as an autograft option in primary pediatric ACL reconstruction has not been well studied. Case series of pediatric patients undergoing quadriceps tendon autograft primary ACL reconstruction demonstrated: no angular deformities/leg length discrepancies; 92% return-to sport; no re-tears, and no re-operations due to graft failure [41–43]. However, no RCT has examined the efficacy of the quadriceps tendon autograft in primary pediatric ACL reconstruction compared to the "reference-standard" soft-tissue hamstring in this population. In light of its evidence for favourable outcomes in the adult population, and the (albeit limited) evidence showing safety and promise in the pediatric population, clinical equipoise exists for assessing its impact on outcomes in pediatric patients at the index surgery.

Research objectives and hypothesis

The primary research objective is to determine, in skeletally-immature patients (ages 10–18 years), the effect of soft-tissue quadriceps versus hamstring autograft tendon during primary ACL reconstruction on rates of ACL graft failure at 24 months. Secondary objectives include determining the effect of soft-tissue quadriceps versus hamstring autograft tendon during primary ACL reconstruction on the rate of return-to-sport (at any level and to preinjury level) using the Tegner Activity Scale, knee function using the Pediatric-International Knee Documentation Committee Scale (Pedi-IKDC), knee pain using a Visual Analogue Scale (VAS), health-related quality of life (HRQL) and health utility using the EuroQol 5 Dimensions for Youth (EQ-5D-Y-3 L), psychological readiness to return-to-sport using the Anterior Cruciate Ligament Return to Sport after Injury (ACL-RSI) scale, range of motion (ROM) and both anterior-posterior and rotational stability through physical examination (i.e. Lachman and pivot shift tests, respectively), and any other adverse events (including reoperation and contralateral injury) at 24 months. We hypothesize that the quadriceps tendon autograft will result in a lower graft failure rate at 24 months compared with hamstring autograft.

Methods

This is a parallel, international, multi-centre, blinded RCT of 352 skeletally-immature (at the time of injury) patients (ages 10–18 years, inclusive) undergoing primary ACL reconstruction to compare the effect of autograft tendon choice (i.e. hamstring versus soft-tissue quadriceps) on the rates of ACL graft failure, return-to-sport, knee function, knee pain, HRQL and health utility, psychological factors, ROM and stability, and any other adverse events at 24 months. Follow-up will occur at 6 weeks, 6 months, 12 months, 18 months and 24 months postoperatively. The trial will be conducted at multiple, international academic and community hospitals that have the expertise to treat the pediatric population undergoing ACL reconstruction. Ethics approval for this trial was granted by the Hamilton Integrated Research Ethics Board (Version 5.0, 11-Feb-2025, HiREB #7571).

Participant selection

Eligibility criteria

The inclusion criteria are: (1) patients aged 10–18 years [40]; (2) history, physical exam, and magnetic resonance imaging (MRI) or arthroscopic image confirmation of ACL insufficiency; (3) suitable for anatomic, single-bundle arthroscopic-assisted ACL reconstruction; (4) plain radiograph or MRI evidence of skeletal immaturity (i.e., any degree of open physes) based on imaging that is closest to the time of injury (and where applicable, correlated with standard bone age left hand posteroanterior radiographs); (5) patients involved in sport (competitive and/or recreational level) prior to injury; (6) patients and parents/guardians speak, read, and understand the language of the clinical site; and (7) patients and parents/guardians provide informed consent and/or assent. Skeletal immaturity may be reconfirmed prior to surgery in the event of delayed surgeries if needed.

The exclusion criteria are: (1) evidence (i.e. radiographic and/or arthroscopic) of International Cartilage Repair Society (ICRS) Cartilage Lesion Classification System Grade 2 (i.e., lesions extending down to 50% of the cartilage depth) and higher osteoarthritis that is symptomatic, requiring treatment other than debridement or microfracture; (2) tibial eminence/spine fractures treated surgically; (3) concomitant collateral, posterior cruciate, and/or cartilage pathology requiring surgical reconstruction and/or advanced restoration techniques (i.e., osteochondral allograft or autograft transfer, matrix-induced autologous chondrocyte implantation, particulate juvenile articular cartilage allograft transplantation); (4) previous ACL reconstruction in the affected knee or contralateral knee; (5) previous distal femur and/or proximal tibial/fibular physeal injury in the affected knee or contralateral knee; (6) allograft or allograft-augmentation, or synthetic augmentation of the ACL reconstruction; (7) biological-augmentation of the ACL reconstruction (i.e. platelet-rich-plasma, fibrin clot, reinforced bioinductive implants, etc.); (8) ACL reconstruction utilizing synthetic grafts; (9) primary ACL repair; (10) patient diagnosed with inflammatory arthropathy; 11) significant medical co-morbidities (requiring daily assistance for activities of daily living); and 12) patient, parent/guardian, and/or clinical investigator believe the patient will have difficulty maintaining follow-up (e.g. patient is planning to move in the next 2 years, unreliable access to transportation to attend follow-up visits, etc.).

Patient screening, recruitment, and randomization

All patients presenting to participating orthopaedic outpatient fracture clinics and/or sports medicine subspecialty clinics between the ages of 10–18 years with an isolated knee injury will be screened by the investigators. Potentially eligible patients (and their parent(s)/guardian(s), if applicable) will be approached by a site-specific Research Coordinator for informed consent (after the patient [and their parent(s)/guardian(s), if applicable] are approached by someone in the patient's circle of care who confirms their interest in the study). To ensure concealment of the treatment allocation, eligible patients will be randomized by the Research Coordinator using the secure, centralized 24-hour online randomization system on REDCap Cloud™. Randomization will be stratified by surgeon and scheduled in random block sizes of 4 and 8.

In the pilot phase, we recruited 100 patients as of September 11, 2024 at 6 centres in Canada and Japan, providing feasibility data and allowing for greater precision for estimates of recruitment and the definitive sample size calculation. We plan to roll these patients into the definitive trial and expand to 15–20 centres across Canada, Brazil, Germany, India, Italy, Japan, Pakistan, Spain, and the USA amongst others.

Interventions

All patients in the study will undergo arthroscopic-assisted, single-bundle, complete, partial, transphyseal, or all-epiphyseal anatomic primary ACL reconstruction at the discretion of the surgeon, considering the individual patient's age, physeal status, and anticipated years of growth remaining to skeletal maturity. All patients will undergo soft-tissue autograft reconstruction, with the control arm using hamstrings (i.e. semitendinosis and/or gracilis) and the treatment arm using all-soft-tissue quadriceps (i.e. full- or partial-thickness) grafts. Grafts will be secured on the femur and tibia according to surgeon preference, given literature demonstrating no clear superior method for graft fixation [44, 45].

All grafts will be prepared according to the surgeon's discretion [46]. Research from our group looking at both quadriceps and hamstring autograft preparation has demonstrated no differences in outcomes based on this parameter [48–51]. Additional research from our group evaluating the utility for the preoperative MRI to predict intraoperative autograft size for ACL reconstruction has demonstrated high correlation for the quadriceps tendon and variable correlation for the combined hamstring tendons [52–54].

The performance of a lateral extra-articular tenodesis (LET) procedure is not prohibited and information regarding this will be collected and analyzed as a covariant in the regression model. Based on trends in adult ACL reconstruction, emerging evidence in the pediatric population with regards to the utility of a LET procedure is increasing. Though preliminary studies examining this procedure in both autograft types used herein in the pediatric and adolescent population are promising, further investigations are needed to validate them and the potential role of LET in this population [55, 56]. The randomization process, blinding, allocation concealment, and intention-to-treat analyses will ensure this covariate is evenly distributed between groups and allow for adequate autograft comparisons.

All aspects of each patient's surgery will be documented including: tourniquet use (i.e. pressure and time); implant methods of graft fixation; tunnel preparation technique (i.e. outside-in, all-inside, accessory antero-medial portal, trans-tibial); knee position for graft fixation (i.e. full extension, 30º); and method for graft tensioning (i.e. manual versus tensiometer, and specific tension amount). In addition, the number of hamstrings (i.e. gracilis and/or semitendinosis), the graft configuration (4-, 5-, 6-stand); the number of strands, the quadriceps autograft width and thickness (i.e. full- or partial-thickness), and the graft diameters and will be recorded.

Surgeon expertise and perioperative care

At a minimum, participating surgeons must have prior dedicated sports medicine training (e.g. fellowship), have been in practice for at least 2 years, and have some research experience working with a Methods Centre. All subjects in both groups will undergo standardized postoperative care in terms of the use of bracing and crutch support, advancement of weightbearing status, and physiotherapy progression for ROM and strengthening [51, 57]. A standardized physiotherapy protocol will be used, whereby patients are provided a pocket protocol explaining recommended strength and conditioning exercises at various time points. The protocol can be completed both at home (for patients without physiotherapy coverage) or with a physiotherapist, and adherence/method data is collected at each follow-up visit. Any deviations from the standardized protocol will be documented.

Study outcomes

Patients will be evaluated at baseline (pre-surgery) and at 6 weeks, 6 months, 12 months, 18 months and 24 months post-surgery. There are visits that should be completed in-person and some that may be completed remotely (Table 1; Fig. 1).

Primary outcome

The primary outcome is the rate of ACL graft failure between the hamstring and soft-tissue quadriceps autograft tendon groups at 24 months postoperatively. ACL graft failure is a composite outcome defined by MRI confirmation of rupture and/or asymmetrical symptomatic laxity (i.e., positive Lachman and/or positive pivot shift) [58, 59]. The expert Adjudication Committee will evaluate all graft failures, reoperations and other knee-related adverse events through a process of independent review and consensus, guided by decision rules detailed in the Adjudication Charter (drafted using recommendations from the literature, tested for validity and agreement among the adjudicators, and used in the pilot study).

Table 1 Schedule of events

Data Collection	Baseline (pre-op)	Surgery	6 weeks^	6 months	12 months	18 months^	24 months
Screening and informed consent	●						
Enrolment data (Demographics)	●						
Randomization form	●						
Baseline x-ray/MRI (to confirm skeletally-immature)	●						
Surgical form		●					
Chondral defects form*		●					
Meniscal injury form*		●					
Follow-up form			●	●	●	●	●
Physical assessment form (blinded assessment)	●				●		●
Tegner activity scale	●		●	●	●	●	●
Pedi-IKDC	●		●	●	●	●	●
EQ-5D-Y-3 L	●		●	●	●	●	●
ACL-RSI	●		●	●	●	●	●
Adverse event form	*		*	*	*	*	*

* = As applicable ^Remote visit permitted

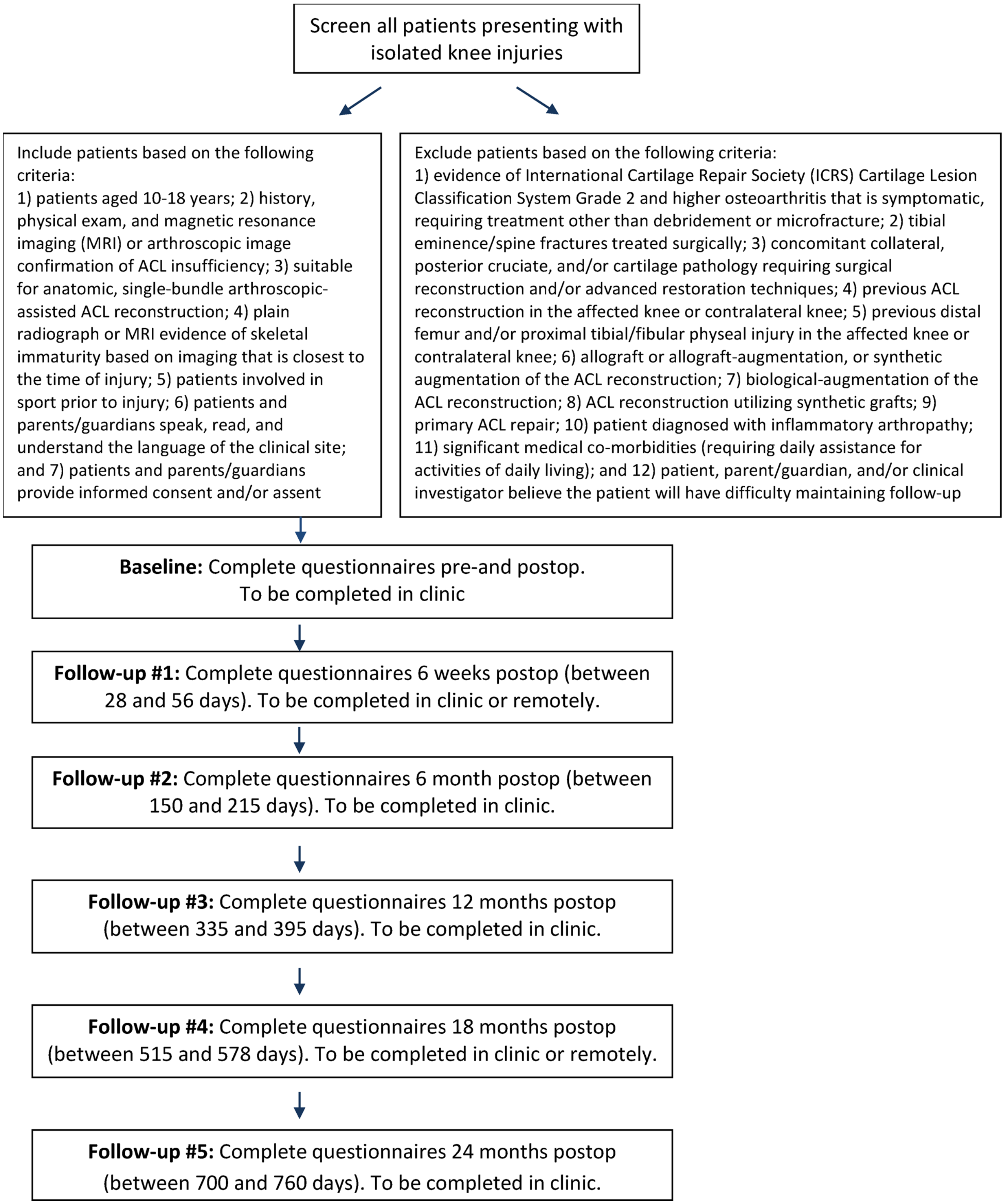

Fig. 1 SQUASH process overview

Secondary outcomes

1. Rate of return-to-sport (at any level and to preinjury level) measured using the Tegner Activity Scale [60].
2. Patient-reported knee function measured using the Pedi-IKDC [61].
3. Knee pain using the VAS [62].
4. HRQL and health utility measured using the EQ-5D-Y-3 L [63].

5. Psychological readiness to return-to-sport using the ACL-RSI [64].
6. ROM (measured using a goniometer) and both anterior-posterior and rotational stability (measured using the Lachman and pivot shift tests [65.66], respectively).
7. Adverse events (including reoperation and contralateral injury).

Return-to-sport and the level of sport will be defined using the Tegner Activity Scale, a standardized method for grading sport activities [60]. This scale ranges from 0 to 10, where 0 is sick leave/disability (due to the knee) and 10 corresponds to play at the national/international level. A score of 6 or greater can only be achieved if participating at a minimum recreational level. This measure demonstrates high test-retest reliability (intra-class coefficient 0.80); has acceptable floor/ceiling effects for patients with ACL tears; is responsive to changes at the study time points, and has been validated in pediatric athletes [60, 67]. For this trial, we will define "return-to-sport" as a "return to unrestricted training at least twice per week, including running at least 30 minutes at least twice per week [47]." In addition, the date the surgeon approved the patient to return-to-play will be collected on the follow-up forms.

The Pedi-IKDC will be used to assess overall knee function. It is widely used and validated for pediatric patients to assess knee symptoms, sport participation, as well as function [61]. This measure has demonstrated excellent test-retest reliability (intra-class coefficient of 0.9) [68], and is valid, reliable, and responsive [69] Specifically, this instrument was referenced by the 2018 International Olympic Committee (IOC) Consensus Statement as the patient-reported outcome measure to assess self-reported knee function [32].

The VAS has been validated in ACL patients and is one of the most commonly used pain rating scores both in research and clinical practice as it is an easy-to-use, validated, one-dimensional scale that does not require verbal or reading skills [70–74].

The EQ-5D is a standardized, utility-based instrument for use as a measure of HRQL and health utility [75]. Similar to the adult version, the EQ-5D-Y-3 L comprises 5 questions on mobility, self-care, usual activities, pain/discomfort, and anxiety/depression [76]. The EQ-5D-Y-3 L has been extensively validated in international pediatric populations [77–80]].

Psychological predictors, specifically psychological readiness to return-to-sport, will be evaluated using the ACL-RSI, which has been shown to be reliable and responsive at the group (study) level in younger populations across all sexes [65, 81, 82]. The ACL-RSI scale has 3 domains: risk appraisal, confidence, and emotions. Greater psychological preparation is indicated by higher scores [66].

ROM will be measured in the flexion-extension (i.e. sagittal) plane using a goniometer [82, 84]. Anterior-posterior stability will be measured using the Lachman test [84], given its non-invasive nature, high sensitivity, superiority over the anterior drawer test, and by virtue of being the most reliable clinical method to assess ACL integrity [65, 66, 82–85]. It will be performed in the supine position, with the extremity in 20–30° of knee flexion, as is commonly described [86, 87]. Similarly, rotational stability will be assessed using the standardized pivot shift test [66, 79], performed in 30° of knee flexion [81]. Both tests are standard, commonly used clinical measures which will increase the applicability and generalizability of the results. Both tests will be completed through physical examination by an independent outcome assessor (i.e., not the treating physician) who will be blinded to the treatment allocation by covering both potential graft donor incision sites with bandages, and who will have specific training in musculoskeletal physical examination and clinical/radiological assessment (e.g., fellow, resident, trainee).

The Adjudication Committee will evaluate all knee-related adverse events including but not limited to, reoperations, contralateral injury, and distal femoral and/or proximal tibial/fibular physeal injury causing growth arrest and/or angular deformity (based on 3-foot standing hip-to-ankle plain radiographs ordered as per standard of care). An adverse event is defined as any symptom, sign, illness, or experience that develops or worsens in severity during the course of the trial.

Sample size calculation

Based on the current literature and aggregate data from the pilot study, we conservatively assume the primary outcome event rate (i.e. ACL graft failure) to be 16% and 6% in the hamstring and quadriceps autograft groups, respectively, at 2 years [88, 89]. To achieve 80% power for a two-sided test at alpha = 0.05, we will require a sample size of 153 in each group. Assuming a conservative loss to follow-up rate of 10% and a crossover rate of 5%, the overall required sample size is 352 subjects (176 per group).

It is important to note that this sample size will also allow for adequate statistical power for our secondary outcomes. To achieve 80% study power using a two-sided type I error rate of 1% to account for multiple comparisons, we will require a total sample size of 192 patients for the Tegner Activity Scale (minimal clinically important difference, MCID = 1; standard deviation, SD = 2) [90, 91], 64 patients for the Pedi-IKDC (MCID = 12, SD

13.5) [92], and 192 patients for the EQ-5D-Y-3 L (MCID = 0.1, SD = 0.2) [93].

Statistical plan

Primary analyses

The baseline characteristics of the patients will be summarized by group and reported as a mean (SD) or median (first quartile, third quartile) for continuous variables and count (percent) for categorical variables. We will adopt the intention-to-treat principle for all analyses—that is, patients will be retained in their randomized groups. To evaluate the effect of hamstring versus quadriceps autograft on rates of graft failure, we will use a Cox proportional hazards model. We will report the treatment effects as hazard ratios and 95% CIs with p-values, where statistical significance will be set at alpha = 0.05. Patients who are lost to follow-up will be censored at the time of their last clinic visit, which will account for missing data due to losses to follow-up.

Secondary analyses

We will estimate the effect of the autograft groups on time return-to-sport using a Cox proportional hazards model, reporting hazard ratios and 95% confidence intervals (CIs). Knee function (Pedi-IKDC), pain (VAS), HRQL and utility (EQ-5D-Y-3 L), ROM and stability at 24 months will be evaluated using independent t-tests, reporting mean differences and 95% CIs. We will also estimate the effect of the treatment groups on adverse events at 24 months using a chi-squared test, presented as an odds ratio with a 95% CI. The criterion for statistical significance will be set at alpha = 0.05 and we will use multiple imputation to account for any missing data (in our continuous and dichotomous outcomes) for all secondary analyses.

Economic analyses

We plan to conduct an economic analysis alongside this trial from the societal perspective. Specifically, the EQ-5D-3Y-L will be used to calculate quality-adjusted life years (QALYs) [88]. We will measure direct costs (surgery, healthcare professionals, hospitalization, diagnosis and procedures, and medications) using the information collected in the case report forms (CRFs), supplemented by participating hospital billing databases. Costs will be estimated in the year the trial is completed and adjusted for inflation and/or purchasing-power parity across participating countries. Given the target patient population is not at work age, indirect costs on productivity loss will not be relevant. Cost effectiveness will be assessed using the incremental cost per QALY gained by comparing the hamstring and quadriceps autograft groups. For the economic analysis, 95% CIs will be estimated using the non-parametric bootstrapping approach by randomly sampling with replacement from individual patient data. Cost effectiveness acceptability curves will be used to present the probability of soft-tissue quadriceps being cost effective over a wide range of willingness to pay thresholds.

Sensitivity analyses

Although we are stratifying randomization by surgeon, we will perform a sensitivity analysis for centre effects, where we will redo the primary analysis including the clinical centre variable in the model. We will also perform a sensitivity analysis for missing data where we only include complete cases. We may also complete adjusted analyses to address any residual baseline imbalance between groups (if found).

Subgroup analyses

We will evaluate body mass index, sex, gender, age, skeletal age, postoperative physiotherapy (i.e., based on adherence to the standardized protocol, involving a physiotherapist or not, and duration), and reconstruction technique as subgroups for our primary outcome. We will add a main effect for each variable and the treatment by subgroup interaction to our primary model described above to assess whether the magnitude of the treatment effect for the primary outcome is significantly different between subgroups.

Data management

The CRFs are the primary data collection tool for the study. An Electronic Data Capture system (REDCap Cloud™) will be used to submit data to the Methods Centre located at McMaster University. Upon receipt of the data, the personnel at the Methods Centre will make a visual check of the data and query all missing, implausible, and inconsistent data.

Data safety monitoring committee

The purpose of the Data Safety Monitoring Committee (DSMC) is to advise the trial investigators regarding the continuing safety of the study participants. The DSMC is comprised of an independent clinical epidemiologist, orthopaedic surgeon, and sports medicine physician who review trial data at regular intervals to ensure participant safety. All members are independent of the trial investigators and have neither financial nor scientific conflicts of interest with the trial.

Ethical considerations

All patients included in the trial will sign a site-specific, Ethics Board-approved consent form that describes this study and provides sufficient information for patients to make an informed decision about their participation. All participating centres must obtain Ethics Board approval from their institution for the study protocol, consent form, CRFs, and any additional protocol amendments prior to enrolment. Any protocol amendments will be communicated to the site investigators, the Ethics Board, trial participants, and trial registries as necessary.

Information about study patients will be kept confidential and will be managed in accordance with the following rules: (1) all study-related information is stored securely at the clinical site, (2) all study patient information is stored in a locked area and is accessible only to study personnel, (3) all CRFs are identified only by a coded patient number, (4) all records that contain patient names, or other identifying information, are stored separately from the study records that are identified only by the coded patient number, and (5) all local and central databases are password protected and access is strictly monitored.

Discussion

The rationale for this trial is based on the: (1) exponential rise in ACL injuries in the pediatric population [4]; (2) high economic burden of ACL injuries (exceeding $1 Billion US annually) [1–3]; (3) high surgical failure rates (10–25%), with this particular population being 15 times more likely to re-injure their graft within the first postoperative year [17–20]; (4) promising results of studies evaluating the quadriceps autograft in an adult population and the limited, lower-quality evidence suggesting safety in the pediatric population [94–99]; (5) void in the literature in terms of high methodological quality studies, specifically RCTs, involving a skeletally-immature population; (6) similar operative equipment, surgical time, and rehabilitation protocols across both autografts, thereby allowing for the possibility of discovering a superior surgical option with no additional costs to the healthcare system; and (7) international support and feasibility for a definitive trial based upon the highly successful 100-patient pilot trial (100% enrollment complete).

This trial is the first RCT evaluating all-soft-tissue tendon autografts in a skeletally-immature population at multiple, international sites to ensure the results are generalizable across a global pediatric population. This trial will overcome many of the limitations and associated biases of the current literature. The trial sample size ($N = 352$) was calculated based on pilot study data to ensure there will be sufficient statistical power to detect differences across several patient important outcomes, including graft failure rates, knee function and pain, and quality of life. Our pilot has also demonstrated feasibility through our ability to recruit patients efficiently; investigator compliance with key aspects of the protocol; maintenance of data quality; maintenance of high follow-up rates (> 90%); and the research team's ability to organize and coordinate trial procedures in a multi-national trial. Furthermore, the trial has methodological safeguards including the use of a centralized randomization system; blinding patients, outcome assessors, and data analysts; standardization and documentation of perioperative care; the use of strategies to limit loss to follow-up; and adjudication of trial events by an independent Adjudication Committee.

As with any surgical RCT, a limitation of the trial is that surgeons cannot be blinded to the treatment allocation. We have put protocols in place to limit possible inconsistencies across sites by standardizing and monitoring critical aspects of operative and perioperative care, including stratifying randomization by surgeon, prescribing a minimum level of expertise for all participating surgeons, and standardizing the postoperative physiotherapy protocol. The trial is designed to be pragmatic and generalizable globally. As a result, all grafts are prepared according to the surgeon's discretion. Prior research has demonstrated no differences in outcomes based on this parameter [47–50].

The 2018 IOC Consensus Statement has made the study of the efficacy of different primary ACL graft options in the pediatric population a research priority [32]. ACL injuries in the pediatric population are on the rise and it is no longer acceptable to delay reconstruction until skeletal maturity [22]. Moreover, reconstruction procedures in this young, active, often "high-risk" population are met with failure rates approaching 10–25% [22]. The quadriceps autograft has demonstrated on par or better performance from a biologic and biomechanical perspective compared to traditional options in the adult population [6]. It should be noted that with the quadriceps tendon graft, surgeons use the same fixation devices and intraoperative equipment, and similarly, surgical time and rehabilitation protocols do not vary. As such, this innovation has significant potential benefit with no additional costs to patients and/or the health care system. Therefore, this study has the potential to change current approaches to soft-tissue primary ACL reconstruction in pediatric patients – adding with a high level of evidence – an additional, and possibly superior, option to the surgeon repertoire.

Abbreviations

ACL	Anterior cruciate ligament
ACL-RSI	Anterior Cruciate Ligament Return to Sport after Injury scale
BPTB	Bone-patella tendon-bone
CI	Confidence interval
CRF	Case report form
DSMC	Data Safety Monitoring Committee
EQ-5D-Y-3L	EuroQol 5 Dimensions for Youth scale
HiREB	Hamilton Integrated Research Ethics Board
HRQL	Health-related quality of life
ICH	International Council for Harmonization
ICRS	International Cartilage Repair Society Cartilage Lesion Classification System
LET	Lateral extra-articular tenodesis
MCID	Minimally clinically important difference
MRI	Magnetic resonance imaging

Pedi-IKDC	Paediatric-International Knee Documentation Committee scale
QALY	Quality-adjusted life year
RCT	Randomized controlled trial
ROM	Range of motion
SD	Standard deviation
TCPSS	Tri-Council Policy Statement: Ethical Conduct for Research Involving Human
VAS	Visual analogue scale

Supplementary Information

The online version contains supplementary material available at https://doi.org/10.1186/s13018-025-06534-0.

Supplementary Material 1

Supplementary Material 2

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Author contributions

DdS, NS, and ORA contributed to the study conception, data abstraction, data analysis, and manuscript writing. All remaining authors (MLN, KN, SC) were involved in manuscript writing and review.

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Data availability

No datasets were generated or analysed during the current study.

Declarations

Competing interests

The authors declare no competing interests.

Ethics approval and consent to participate

Ethics approval for this trial was granted by the Hamilton Integrated Research Ethics Board (Version 5.0, 11-Feb-2025, HiREB #7571) in accordance with the Tri-Council Policy Statement: Ethical Conduct for Research Involving Human (TCPS2) and Good Clinical Practice Standards set by the International Council for Harmonization (ICH).

Consent for publication

Not applicable.

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References

1. Kim HS, Seon JK, Jo AR. Current trends in anterior cruciate ligament reconstruction. Knee Surg Relat Res. 2013;25(4):165–73.
2. Kaeding CC, Leger-St-Jean B, Magnussen RA. Epidemiology and diagnosis of anterior cruciate ligament injuries. Clin Sports Med. 2017;36(01):1–8.
3. Lynch TS, Parker RD, Patel RM, MOON Group, et al. The impact of the multicenter orthopaedic outcomes network (MOON) research on anterior cruciate ligament reconstruction and orthopaedic practice. J Am Acad Orthop Surg. 2015;23(03):154–63.
4. Maffulli N, Del Buono A. Anterior cruciate ligament tears in children. Surgeon. 2013;11(2):59–62.
5. Maffulli N, Longo UG, Gougoulias N, Loppini M, Denaro V. Long-term health outcomes of youth sports injuries. Br J Sports Med. 2010;44(1):21–5.
6. Fabricant PD, Kocher MS. Anterior cruciate ligament injuries in children and adolescents. Orthop Clin North Am. 2016;47(4):777–88.
7. Beck NA, Lawrence JTR, Nordin JD, DeFor TA, Tompkins M. ACL tears in school-aged children and adolescents over 20 years. Pediatrics. 2017;139(3):e20161877.
8. Frobell R, Roos E, Roos H, Ranstam J, Lohmander L. A randomized trial of treatment for acute anterior cruciate ligament tears. N Engl J Med. 2010;363:331–42.
9. Anderson AF, Anderson CN. Correlation of meniscal and articular cartilage injuries in children and adolescents with timing of anterior cruciate ligament reconstruction. Am J Sports Med. 2015;43:275–81.
10. Lawrence JT, Argawal N, ganley TJ. Degeneration of the knee joint in skeletally immature patients with a diagnosis of an anterior cruciate ligament tear: is there harm in delay of treatment? Am J Sports Med. 2011;39:2582–7.
11. Migliorini F, Pilone M, Schäfer L, et al. Allograft versus autograft ACL reconstruction in skeletally immature patients: a systematic review. Br Med Bull. 2025;153(1):ldae020.
12. Migliorini F, Cocconi F, Schäfer L, et al. Bone-patellar tendon-bone, hamstring, and quadriceps tendon autografts for anterior cruciate ligament reconstruction in skeletally immature patients: a systematic review. Br Med Bull. 2024;152(1):16–27.
13. Newman JT, Carry PM, Terhune EB. Factors predictive of concomitant injuries among children and adolescents undergoing anterior cruciate ligament surgery. Am J Sports Med. 2015;43:282–8. https://doi.org/10.1177/0363546514562168.
14. Bach BR Jr. Revision anterior cruciate ligament surgery. Arthroscopy. 2003;19(S1):14–29.
15. Migliorini F, Cocconi F, Schäfer L, Memminger MK, Giorgino R, Maffulli N. Anterior cruciate ligament reconstruction in skeletally immature patients is effective: A systematic review. Knee Surg Sports Traumatol Arthrosc. 2024;32(2):418–31.
16. Baer GS, Harner CD. Clinical outcomes of allograft versus autograft in anterior cruciate ligament reconstruction. Clin Sports Med. 2007;26(04):661–81.

17. Engelman GH, Carry PM, Hitt KG, Polousky JD, Vidal AF. Comparison of allograft versus autograft anterior cruciate ligament reconstruction graft survival in an active adolescent cohort. Am J Sports Med. 2014;42(10):2311–8.
18. Ho B, Edmonds EW, Chambers HG, Bastrom TP, Pennock AT. Risk factors for early ACL reconstruction failure in paediatric and adolescent patients: a review of 561 cases. J Pediatr Orthop. 2018;38(7):388–92.
19. DeFrancesco CJ, Storey EP, Flynn JM, Ganley TJ. Pediatric ACL reconstruction and return to the operating room: revision is less than half of the story. J Pediatr Orthop. 2019;39(10):516–20.
20. Graziano J, Chiaia T, de Mille P, Nawabi DH, Green DW, Cordasco FA. Return to sport for skeletally immature athletes after ACL reconstruction: preventing a second injury using a quality of movement assessment and quantitative measures to address modifiable risk factors. Orthop J Sports Med. 2017;5(4):2325967117700599.
21. Senorski E, Seil R, Svantesson E, Feller J, Webster K, Engebretsen L, Spinderl K, Siebold R, Karlsson J, Samuelsson. I never made it to the pros… return to sport and becoming an elite athlete after paediatric and adolescent anterior cruciate ligament injury – Current evidence and future directions. Knee Surg Sports Traumatol Arthrosc. 2018;26(4):1011–8.
22. Peterson DC, Ayeni OR. Paediatric anterior cruciate ligament reconstruction outcomes. Curr Rev Musculoskelet Med. 2016;9(4):339–47.
23. Ardern CL, Taylor NF, Feller JA, Webster KE. Fifty-five per cent return to competitive sport following anterior cruciate ligament reconstruction surgery: an updated systematic review and metaanalysis including aspects of physical functioning and contextual factors. Br J Sports Med. 2014;48(21):1543–52.
24. Joreitz R, Lynch A, Rabuck S, Lynch B, Davin S, Irrgang J. Patient-specific and surgery-specific factors that affect return to sport after ACL reconstruction. Int J Sports Phys Ther. 2016;11(02):264–78.
25. Lai CCH, Ardern CL, Feller JA, Webster KE. Eighty-three per cent of elite athletes return to preinjury sport after anterior cruciate ligament reconstruction: a systematic review with meta-analysis of return to sport rates, graft rupture rates and performance outcomes. Br J Sports Med. 2018;52(02):128–38.
26. Christino MA, Fantry AJ, Vopat BG. Psychological aspects of recovery following anterior cruciate ligament reconstruction. J Am Acad Orthop Surg. 2015;23(08):501–9.
27. Coronado RA, Bird ML, Van Hoy EE, Huston LJ, Spindler KP, Archer KR. Do psychosocial interventions improve rehabilitation outcomes after anterior cruciate ligament reconstruction? A systematic review. Clin Rehabil. 2018;32(03):287–98.
28. Group MARS, Wright RW, Huston LJ, Spindler KP, Dunn WR, Haas AK, Allen CR, Cooper DE, DeBerardino TM, Lantz BB, Mann BJ, Stuart MJ. Descriptive epidemiology of the multicenter ACL revision study (MARS) cohort. Am J Sports Med. 2010;38(10):1979–86.
29. MARS Group. Effect of graft choice on the outcome of revision anterior cruciate ligament reconstruction in the multicenter ACL revision study (MARS) cohort. Am J Sports Med. 2014;42(10):2301–10.
30. Biau DJ, Tournoux C, Katsahian S, Schranz P, Nizard R. ACL reconstruction: a meta-analysis of functional scores. Clin Orthop Relat Res. 2007;458:180–7.
31. Spindler KP, Kuhn JE, Freedman KB, Matthews CE, Dittus RS, Harrell FE Jr. Anterior cruciate ligament reconstruction autograft choice: bone-tendon-bone versus hamstring: does it really matter? A systematic review. Am J Sports Med. 2004;32:1986–95.
32. Migliorini F, Pilone M, Memminger MK, et al. All-epiphyseal anterior cruciate ligament reconstruction yields superior sports performances than the trans-epiphyseal technique in skeletally immature patients: a systematic review. J Orthop Traumatol. 2024;25(1):7.
33. Ardern CL, Ekås G, Grindem H, et al. 2018 international olympic committee consensus statement on prevention, diagnosis and management of paediatric anterior cruciate ligament (ACL) injuries. Knee Surg Sports Traumatol Arthrosc. 2018;26(4):989–1010.
34. Richmond CG, Green DW, Cannamela PC, Martinson D, Shea KG. The distance between the hamstring grafts and the physis and apophysis of the proximal tibia – implications for pediatric ACL reconstruction and physeal arrest. J ISAKOS. 2018;3:318–22.
35. Cooper DE, Deng XH, Burstein AL, Warren RF. The strength of the central third patellar tendon graft. A Biomechanical study. Am J Sports Med. 1993;21(6):818–23.
36. Cohen M, Ferretti M, Quarteiro M. Transphyseal anterior cruciate ligament reconstruction in patients with open physes. Arthroscopy. 2009;25:831–8.
37. Fabricant PD, Kocher MS. Management of ACL injuries in children and adolescents. J Bone Joint Surg Am. 2017;99:600–12.
38. Hui C, Roe J, Ferguson D. Outcome of anatomic transphyseal anterior cruciate ligament reconstruction in Tanner stage 1 and 2 patients with open physes. Am J Sports Med. 2012;40:1093–8.
39. Kocher MS, Smith JT, Zoric BJ, Lee B, Micheli LJ. Transphyseal anterior cruciate ligament reconstruction in skeletally immature pubescent adolescents. J Bone Joint Surg Am. 2007;89:2632–9.
40. Dekker TJ, Godin JA, Dale KM, Garrett WE, Taylor DC, Riboh JC. Return to sport after paediatric anterior cruciate ligament reconstruction and its effect on subsequent anterior cruciate ligament injury. J Bone Joint Surg Am. 2017;99(11):897–904.
41. Puddu G. Method for reconstruction of the anterior cruciate ligament using the semitendinosus tendon. Am J Sports Med. 1980;8(6):402–4.
42. DeAngelis JP, Fulkerson JP. Quadriceps tendon–a reliable alternative for reconstruction of the anterior cruciate ligament. Clin Sports Med. 2007;26(4):587–96.
43. Reinhardt KR, Hetsroni I, Marx RG. Graft selection for anterior cruciate ligament reconstruction: a level I systematic review comparing failure rates and functional outcomes. Orthop Clin North Am. 2010;41(2):249–62.
44. Sofu H, Sahin V, Gürsu S, Yıldırım T, Issın A, Ordueri M. Use of quadriceps tendon versus hamstring tendon autograft for arthroscopic anterior cruciate ligament reconstruction: a comparative analysis of clinical results. Eklem Hastalik Cerrahisi. 2013;24(3):139–43.
45. Haner M, Bierke S, Petersen W. Anterior cruciate ligament revision surgery: ipsilateral quadriceps versus contralateral semitendinosus-gracilis autografts. Arthroscopy. 2016;32(11):2308–17.
46. Crum RJ, de Sa D, Kanakamedala AC, Obioha OA, Lesniak BP, Musahl V. Aperture and suspensory fixation equally efficacious for quadriceps tendon graft fixation in primary ACL reconstruction: a systematic review. J Knee Surg. 2020;33(7):704–21.
47. Lubowitz JH, Schwartzberg R, Smith P. Cortical suspensory button versus aperture interference screw fixation for knee anterior cruciate ligament soft-tissue allograft: a prospective, randomized controlled trial. Arthroscopy. 2015;31(9):1733–9.
48. Mariscalco MW, Flanigan DC, Mitchell J, Pedroza AD, Jones MH, Andrish JT, Parker RD, Kaeding CC, Magnussen RA. The influence of hamstring autograft size on patient-reported outcomes and risk of revision after anterior cruciate ligament reconstruction: a multicenter orthopaedic outcomes network (MOON) cohort study. Arthroscopy. 2013;29(12):1948–53.
49. Vivekanantha P, Nedaie S, Grzela P, Cohen D, Hoshino Y, Nagai K, Kay J, de Sa D. Though five-strand hamstring autografts demonstrate greater graft diameter, postoperative outcomes are equivocal to four-strand hamstring autograft preparations: a systematic review and meta-analysis of level I and II studies. Knee Surg Sports Traumatol Arthrosc. 2023;31(10):4437–47.
50. Sun B, Lee B, Grad J, Cohen D, Abouali J, Tapasvi S, Maniar A, de Sa D. Anterior cruciate ligament reconstruction with six and eight-strand hamstring tendon autografts produces adequate graft dimensions and functional outcomes: a systematic review. Knee Surg Sports Traumatol Arthrosc. 2024 https://doi.org/10.1002/ksa.12556.
51. Kanakamedala AC, de Sa D, Obioha OA, Arakgi ME, Schmidt PB, Lesniak BP, Musahl V. No difference between full thickness and partial thickness quadriceps tendon autografts in anterior cruciate ligament reconstruction: a systematic review. Knee Surg Sports Traumatol Arthrosc. 2019;27(1):105–16.
52. Crum RJ, Kay J, Lesniak BP, Getgood A, Musahl V, de Sa D. Bone versus all soft tissue quadriceps tendon autografts for anterior cruciate ligament reconstruction: a systematic review. Arthroscopy. 2021;37(3):1040–52.
53. Vivekanantha P, Sun B, Parasu N, de Sa D. Magnetic resonance imaging can predict hamstring or quadriceps tendon autograft diameter in pediatric or adolescent anterior cruciate ligament reconstruction: a systematic review. J Pediatr Orthop. 2025;45(2):e112–8.
54. Dworsky-Fried J, Hadwen A, Bernardini L, Vivekanantha P, Grassi A, Ollivier M, de Sa D. Quadriceps tendon autograft diameters are routinely above 8 mm, and preoperative size estimation before anterior cruciate ligament reconstruction may not be necessary for this graft type: a systematic review. Knee Surg Sports Traumatol Arthrosc. 2024 https://doi.org/10.1002/ksa.12558.
55. Monaco E, Carrozzo A, Saithna A, et al. Isolated ACL reconstruction versus ACL reconstruction combined with lateral Extra-articular tenodesis: A comparative study of clinical outcomes in adolescent patients. Am J Sports Med. 2022;50(12):3244–55.
56. Carrozzo A, Monaco E, Saithna A, et al. Clinical outcomes of combined anterior cruciate ligament reconstruction and lateral Extra-articular tenodesis procedures in skeletally immature patients: A systematic review from the SANTI study group. J Pediatr Orthop. 2023;43(1):24–30.

57. Agarwal S, de Sa D, Peterson DC, Parmar D, Simunovic N, Ogilvie R, Musahl V, Ayeni OR. Can preoperative magnetic resonance imaging predict intraoperative autograft size for anterior cruciate ligament reconstruction? A systematic review. J Knee Surg. 2019;32(7):649–58.
58. Zhang K, Beshay T, Murphy B, Sheean A, de Sa D. Quadriceps tendon anterior cruciate ligament reconstruction: a systematic review of postoperative rehabilitation and complication profiles. Arthroscopy. 2022;38(6):2062–e20721.
59. Wasserstein D, Sheth U, Cabrera A, Spindler KP. A systematic review of failed anterior cruciate ligament reconstruction with autograft compared with allograft in young patients. Sports Health. 2015;7(3):207–16.
60. Clinicaltrials.gov. STABILITY 2: anterior cruciate ligament reconstruction +/- lateral tenodesis with patellar vs quad tendon (STABILITY 2). NCT03935750. https://clinicaltrials.gov/study/NCT03935750
61. Briggs KK, Lysholm J, Tegner Y, Rodkey WG, Kocher MS, Steadman JR. The reliability, validity, and responsiveness of the Lysholm score and Tegner activity scale for anterior cruciate ligament injuries of the knee: 25 years later. Am J Sports Med. 2009;37(5):890–7.
62. Nasreddine AY, Connell PL, Kalish LA, Nelson S, Iversen MD, Anderson AF, Kocher MS. The paediatric international knee Documentation committee (Pedi-IKDC) subjective knee evaluation form: normative data. Am J Sports Med. 2017;45(3):527–34.
63. Price DD, McGrath PA, Rafii A, Buckingham B. The validation of visual analogue scales as ratio scale measures for chronic and experimental pain. Pain. 1983;17:45–56.
64. Wille N, Badia X, Bonsel G, Burström K, Cavrini G, Devlin N, Egmar AC, Greiner W, Gusi N, Herdman M, Jelsma J, Kind P, Scalone L, Ravens-Sieberer U. Development of the EQ-5D-Y: a child-friendly version of the EQ-5D. Qual Life Res. 2010;19(6):875–86.
65. Webster KE, Feller JA, Lambros C. Development and preliminary validation of a scale to measure the psychological impact of returning to sport following anterior cruciate ligament reconstruction surgery. Phys Ther Sport. 2008;9(1):9–15.
66. Hoshino Y, Araujo P, Ahlden M, Moore CG, Kuroda R, Zaffagnini S, Karlsson J, Fu FH, Musahl V. Standardized Pivot shift test improves measurement accuracy. Knee Surg Sports Traumatol Arthrosc. 2012;20(4):732–6.
67. Kim SJ, Kim HK. Reliability of the anterior drawer test, the Pivot shift test, and the Lachman test. Clin Orthop Relat Res. 1995;(317):237–42.
68. Placella G, Bartoli M, Peruzzi M, Speziali A, Pace V, Cerulli G. Return to sport activity after anterior cruciate ligament reconstruction in skeletally immature athletes with manual drilling original all inside reconstruction at 8 years follow-up. Acta Orthop Traumatol Turc. 2016;50(6):635–8.
69. Jacobsen JS, Knudsen P, Fynbo C, Rolving N, Warming S. Reproducibility and responsiveness of a Danish Pedi-IKDC subjective knee form for children with knee disorders. Scand J Med Sci Sports. 2016;26(12):1408–14.
70. Dietvorst M, Reijman M, van Groningen B, Janssen RPA. PROMs in paediatric knee ligament injury: use the Pedi-IKDC and avoid using adult proms. Knee Surg Sports Traumatol Arthrosc. 2019;27(6):1965–73.
71. Jensen MP, Karoly P. The measurement of clinical pain intensity: a comparison of six methods. Pain. 1986;27:117–26.
72. Collins S, Moore A, McQuay H. The visual analog pain intensity scale: what is moderate pain in millimeters? Pain. 1997;72:95–7.
73. Ho K, Spence J, Murphy M. Review of pain measurement tools. Ann Emerg Med. 1996;27:427–31.
74. Höher J, Münster A, Klein J, et al. Validation and application of a subjective knee questionnaire. Knee Surg Sports Traumatol Arthrosc. 1995;3(1):26–33.
75. Lutz C, Baverel L, Colombet P, et al. French arthroscopic Society. Pain after out-patient vs. in-patient ACL reconstruction: French prospective study of 1076 patients. Orthop Traumatol Surg Res. 2016;102(8S):S265–70.
76. EuroQol Group. The EQ-5D. Available at: https://euroqol.org/information-and-support/
77. EuroQol Group. The EQ-5D-Y-3L. Available at: https://euroqol.org/information-and-support/euroqol-instruments/eq-5d-y-3l/
78. Ravens-Sieberer U, Wille N, Badia X, Bonsel G, Burström K, Cavrini G, Devlin N, Egmar AC, Gusi N, Herdman M, Jelsma J, Kind P, Olivares PR, Scalone L, Greiner W. Feasibility, reliability, and validity of the EQ-5D-Y: results from a multinational study. Qual Life Res. 2010;19(6):887–97.
79. Ramos-Goñi JM, Oppe M, Stolk E, Shah K, Kreimeier S, Rivero-Arias O, Devlin N. International valuation protocol for the EQ-5D-Y-3L. PharmacoEconomics. 2020;38(7):653–63.
80. Kreimeier S, Greiner W. EQ-5D-Y as a health-related quality of life instrument for children and adolescents: the instrument's characteristics, development, current use, and challenges of developing its value set. Value Health. 2019;22(1):31–7.
81. Sell TC, Zerega R, King V, Reiter CR, Wrona H, Bullock GS, Mills N, Räisänen A, Ledbetter L, Collins GS, Kvist J, Filbay SR, Losciale JM. Anterior cruciate ligament return to sport after injury scale (ACL-RSI) scores over time after anterior cruciate ligament reconstruction: a systematic review with meta-analysis. Sports Med Open. 2024;10(1):49.
82. Webster KE, Feller JA. Evaluation of the responsiveness of the anterior cruciate ligament return to sport after injury (ACL-RSI) scale. Orthop J Sports Med. 2021;9(8):23259671211031240.
83. Lam MH, Fong DT, Yung PSh, Ho EP, Chan WY, Chan KM. Knee stability assessment on anterior cruciate ligament injury: clinical and Biomechanical approaches. Sports Med Arthrosc Rehabil Ther Technol. 2009;1(1):20.
84. Peters PG, Herbenick MA, Anloague PA, Markert RJ, Rubino LJ 3rd. Knee range of motion: reliability and agreement of 3 measurement methods. Am J Orthop (Belle Mead NJ). 2011;40(12):E249–52.
85. Gurtler RA, Stine R, Torg JS. Lachman test revisited. Contemp Orthop. 1990;20(2):145–54.
86. Mitsou A, Vallianatos P. Clinical diagnosis of ruptures of the anterior cruciate ligament: a comparison between the Lachman test and the anterior drawer sign. Injury. 1988;19(6):427–8.
87. van Eck CF, Fu FH, Poolman RW, Kerkhoffs GM. Methods to diagnose acute anterior cruciate ligament rupture: a meta-analysis of physical examinations with and without anaesthesia. Knee Surg Sports Traumatol Arthrosc. 2013;21(8):1895–903.
88. Arilla FV, Yeung M, Bell K, Rahnemai-Azar AA, Rothrauff BB, Fu FH, Debski RE, Ayeni OR, Musahl V. Experimental execution of the simulated pivot-shift test: a systematic review of techniques. Arthroscopy. 2015;31(12):2445–54.
89. Jackson GR, Mameri ES, Tuthill T, Wessels M, Asif S, Sugrañes J, Batra AK, McCormick JR, Obioha OA, Kaplan DJ, Knapik DM, Verma NN, Chahla J. Adverse events and complications after primary ACL reconstruction with quadriceps tendon autograft: a systematic review. Orthop J Sports Med. 2023;11(12):23259671231199728.
90. Lord L, Cristiani R, Edman G, Forssblad M, Stålman A. One sixth of primary anterior cruciate ligament reconstructions May undergo reoperation due to complications or new injuries within 2 years. Knee Surg Sports Traumatol Arthrosc. 2020;28(8):2478–85.
91. Weng CJ, Yeh WL, Hsu KY, Chiu CH, Chang SS, Chen AC, Chan YS. Clinical and functional outcomes of anterior cruciate ligament reconstruction with autologous hamstring tendon in patients aged 50 years or older. Arthroscopy. 2020;36(2):558–62.
92. Runer A, Suter A, Roberti di Sarsina T, Jucho L, Gföller P, Csapo R, Hoser C, Fink C. Quadriceps tendon autograft for primary anterior cruciate ligament reconstruction show comparable clinical, functional, and patient-reported outcome measures, but lower donor-site morbidity compared with hamstring tendon autograft: a matched-pairs study with a mean follow-up of 6.5 years. J ISAKOS. 2023;8(2):60–7.
93. Hiemstra LA, Page JL, Kerslake S. Patient-reported outcome measures for patellofemoral instability: a critical review. Curr Rev Musculoskelet Med. 2019;12(2):124–37.
94. Bhandari M, Devereaux PJ, Einhorn TA, Thabane L, Schemitsch EH, Koval KJ, Frihagen F, Poolman RW, Tetsworth K, Guerra-Farfán E, Madden K, Sprague S, Guyatt G. HEALTH Investigators. Hip fracture evaluation with alternatives of total hip arthroplasty versus hemiarthroplasty (HEALTH): protocol for a multicentre randomised trial. BMJ Open. 2015;5(2):e006263.
95. Whitehead SJ, Ali S. Health outcomes in economic evaluation: the QALY and utilities. Br Med Bull. 2010;96:5–21.
96. Mulford JS, Hutchinson SE, Hang JR. Outcomes for primary anterior cruciate reconstruction with the quadriceps autograft: a systematic review. Knee Surg Sports Traumatol Arthrosc. 2013;21(8):1882–8.
97. Albright J, Lepon AK, Mayer S. Anterior cruciate ligament reconstruction in pediatric and adolescent patients using quadriceps tendon autograft. Sports Med Arthrosc Rev. 2016;24(4):159–69.
98. Mauch C, Arnold MP, Wirries A, et al. Anterior cruciate ligament reconstruction using quadriceps tendon autograft for adolescents with open physes - a technical note. Sports Med Arthrosc Rehabil Ther Technol. 2011;3(1):7.

99. Pennock AT, Johnson KP, Chambers HG, et al. Quadriceps tendon autografts have a lower early graft failure rate than hamstring tendon autografts when performing transphyseal ACL reconstructions. Orthop J Sports Med. 2019;7(7 suppl5):2325967119S00294.

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